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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025
THIRD SEMESTER
DISCIPLINE SPECIFIC CORE COURSE (DSC) - MG3DSCZGY201,
DISCIPLINE SPECIFIC ELECTIVE (DSE)- MG3DSEZIM 200-
ANIMAL DIVERSITY NON CHORDATA-II
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30Minutes **Maximum Marks: 50**

Part A

Multiple Choice Questions

Answer all questions. Each question carries **1** mark

1. The duration of life cycle of silkworm is **[K]/[CO1]**
   a) 6-8 weeks
2. The compound eyes of prawns exhibit which type of vision in dim light? **[U]/[CO2]**
   b)Superposition image vision
3. Write the characteristic shell shape of Scaphopoda **[K]/[CO3]**
   d) tusk shaped
4. Discuss the significance of Branchiopoda in aquatic ecosystems. **[U]/[CO1]**
   b) primary consumers in freshwater food chain
5. The gills of Fenneropenaeus indicus are located in which part of the body? **[K]/[CO2]**
   b) Branchial chamber under the carapace
6. Name the marine burrowing organism lived in mucus lined burrows **[K]/[CO3]**
   b) Sipunculus
7. The first larval form of Sacculina is? **[K]/[CO1]**
   a) Nauplius
8. The ventral plate of a prawn’s body segment is called the **[K]/[CO2]**
   b)Sternum
9. Tube feet in Echinodermata are used for: **[U]/[CO3]**
   a) Locomotion and feeding
10. The body of trilobites is divided into which regions? **[U]/[CO1]**
    b) Cephalon, thorax, pygidium
11. A farmer notices reduced crop yield even though there are no visible pests. On investigation, it is found that pollinator insects have reduced in the area. What does this indicate? **[U]/[CO2]**
    c) lack of pollination reduces crop productivity
12. Select the nutritional value of edible mollusc **[U]/[CO3]**
    a) High protein and low fat

Part B

Short Answer Questions

Answer any 6 questions. Each question carries **3** marks. **[3×6=18]**

13. Comment on any four salient features of Monoplacophora. [K]/[CO3]

i) Fragile and single piece shell ii)Bilaterally symmetrical and segmented body with serially repeated external gills,nephridia iii)Serially arranged eight pairs of retractor muscles in the foot iv)Development with trocophore larva ***(Any four points)***

14. Categorize the gills in prawn. [U]/[CO2]

Podobranchs (foot gills)- attaches to the coxa of thoracic appaendages.
Arthrobranchs – attached to the junction between thoracic wall and appendages.
Pleurobranchs- attach to the inner wall of the branchial chamber.
(Give full mark if all 3 are attempted).

15. Report one economic importance of Argulus. [U]/[CO1]

Significant negative economic importance in aquaculture. It is an ectoparasite on fishes.

16.Describe why Limulus is called a living fossil. [U]/[CO1]

Similar species were present during the Triassic and Devonian periods. They have not undergone any significant evolutionary change.They are closest living relatives of extinct trilobites.

17. Describe the salient features of Hemichordata. [K]/[CO3]

Marine, wormlike body, soft, fragile, unsegmented, bilateraly symmetrical, triploblastic, enterocoelous, buccal diverticulum, pharyngeal gill slits, circulatory system simple and open, nervous system primitive, sexes separate, body composed of proboscis, collar and trunk.
(Give full mark if any 3 points are attempted)

18. Describe the body symmetry exhibited by adult echinoderms and its evolutionary significance. [U]/[CO3]

Larval forms of echinoderms are bilaterally symmetrical and adults are radially symmetrical. It is believed that echinoderms evolved from bilaterally symmetrical ancestors.
(Liberal valuation)

19. Mention three ecological advantages of larval metamorphosis in prawn. [K]/[CO2]

The reduced competition with adults, enhanced dispersal to new habitats, and specialized adaptations for different environmental conditions. ***(Liberal Valuation)***

20. Name any three larvae of the phylum Mollusca. [K]/[CO3]

Trochophore, veliger and glochidium**.**

Part C

Short Essay Questions

Answer any 4 questions. Each question carries **5** marks. **[5×4=20]**

21.Describe the morphological features of Scolopendra. [K]/[CO1]

Dorsoventrally flattened body consists of head and trunk (15– 100 segments). One pair of legs in each segment. They have a head with a flat cephalic plate, antennae, and venomous forcipules (modified legs) instead of the first pair of legs. ***(Give full marks if any 5points are attempted).***

22. Discuss the structure and function of thoracic appendages in prawns. [U]/[CO2]

Thoracic appendages in prawns include three pairs of maxillipeds and five pairs of pereiopods (walking legs). All are biramous, and have protopodite of two podomeres and endopodite of 5 podomeres and an unsegmented exopodite. Coxa of first 3 thoracic appendages bears epipodite, which is used for respiration. Second maxilliped curved like a question mark. First 3 pairs of walking legs are chelate type and last two are non-chelate type. Maxillipeds are used for Handling and manipulation of food. Pereiopods have walking, grasping, and reproductive or cleaning roles. ***(Diagram not necessary, If diagram of all thoracic appendages drawn,give full mark)***

23. Discuss the ecological and economic importance of Holothuroidea. [U]/[CO3]

Ecological importance- Holothurians are detritivores. They perform crucial ecosystem services by recycling nutrients and maintaining seabed health. The physical mixing and burrowing of sediments by sea cucumbers (bioturbation) improve the oxygenation and permeability of seafloor substrates.
Economic importance- Holothuroidea have been harvested for centuries and are used for producing high-value seafood delicacyand medicinal products. ***(Liberal valuation)***.

24. Write an essay on subphylum Uniramia and its classification with suitable example. [U]/[CO1]

Uniramia is a subphylum of Arthropoda. ***(0.5 Marks).*** Salient features: Body divisible into head and trunk / head, thorax and abdomen; Head bears cephalic appendages, ocelli, compound eyes; Thorax is 3 segmented and bears 3 pairs of walking legs; Abdomen is many segmented; Respiration by tracheae, excretion by malpighian tubules or cephalic maxillary glands. Sexes separate. ***(Any 2 features-2 Marks)***

The subphylum is divided into 5 classes:
Class 1: Chilopoda- eg. Scolopendra
Class 2: Symphyla-eg. Scutigera
Class 3: Diplopoda- eg.Spirostreptus
Class 4: Pauropoda- eg.Pauropus
Class 5: Hexapoda- eg.Bombyxmori
(Classification- 2.5 Marks)

25. Explain the circulatory system in prawn. **[U]/[CO2]**

The prawn has an open circulatory system. The blood vascular system consists of heart, pericardium, arteries, blood lacunae, and haemolymph (haemocyanin). The heart pumps the blood into arteries, which branch out into the body cavity, allowing the blood to bathe the organs directly. From there, the blood returns to the heart via a series of channels and sinuses after passing through the gills to pick up oxygen. Blood carried to gills by afferent branchial channels. Pure blood after oxygenation is carried to the pericardial sinus by efferent branchial channels. From there blood flows to heart through ostia.

26. Write a short essay on excretory system of prawn. **[U]/[CO2]**

The excretory system of a prawn is composed of a pair of antennary glands, also known as green glands, which are located at the base of the antennae. Each gland has 4 parts: end sac, labyrinth, glandular tube, and bladder. A small excretory duct is continued from the bladder and it opens to the exterior by an excretory pore on the coxal segment of antenna. ***(Explanation of all the parts- 5 marks)***